

RUNNING A MERCATURA FULL NODE

Public Testnet • Step-by-Step Guide

Written for first-time users — no blockchain experience required

Start here

A Mercatura full node independently verifies the blockchain and relays valid blocks and transactions to other peers. You do not need to mine, run an Explorer, or expose RPC to operate a useful full node.

This guide is for TESTNET. Use test MCA only. Testnet coins have no monetary value. Mainnet will use a separate release, data directory, chain history and ports. Do not reuse a testnet data directory or configuration as a mainnet wallet.

Choose your path

Path	Best for	Difficulty
A — Windows GUI	Most testers; easiest way to run a normal full node	Easy
B — Ubuntu desktop / terminal	Linux users who want Mercatura Qt or a manually started daemon	Easy-moderate
C — Ubuntu 22.04 always-on daemon	VPS, home server, 24/7 public node	Moderate

What you need

- The official Mercatura Core testnet release for your operating system.
- The published SHA-256 checksum for that release.
- The current testnet bootstrap node address, written as HOST_OR_IP:27778.
- Enough free disk space for the testnet chain and normal operating-system overhead.
- An Internet connection that can stay online while the node synchronizes.

One thing you must replace. Every place this guide shows BOOTSTRAP_HOST, replace it with the official Mercatura public testnet bootstrap hostname or IP announced with the release.

1. What a full node does

A full node downloads block headers and blocks, verifies Mercatura consensus rules itself, maintains chain state, and exchanges data with peers. It does not have to trust an Explorer or another node's opinion about which transactions and blocks are valid.

1.1 Mercatura testnet network facts

Item	Testnet value
Network	Mercatura testnet
P2P port	27778/tcp — public peer-to-peer networking
RPC port	27775/tcp — local/private only
Address HRP	tmca — native testnet addresses begin with tmca...
Configuration file	mercatura.conf
Windows default data directory	C:\Users\<you>\AppData\Local\Mercatura
Linux default data directory	~/.mercatura
Qt application	mercatura-qt / mercatura-qt.exe
Headless daemon	mercaturad
Command-line client	mercatura-cli

1.2 Important safety rules

- Never expose RPC port 27775 to the public Internet. Ordinary users do not need public RPC at all.
- Opening P2P port 27778 is safe and useful when your router/firewall is configured correctly; it allows inbound peers.
- A node with only outbound connections still validates the chain and is useful to you, but it contributes less inbound capacity to the network.
- Do not store valuable mainnet credentials or unrelated private keys on a public testnet VPS.
- Back up any wallet you care about before deleting or replacing its data directory, even on testnet.

1.3 Why you may need a manual bootstrap peer

Mercatura's early public networks intentionally begin without inherited Bitcoin seeds and may have no DNS/fixed seeds yet. That means a fresh node may start with zero peers until it is told about at least one known Mercatura peer. After the first connection, peer address exchange can help it learn additional nodes.

```
addnode=BOOTSTRAP_HOST:27778
```

2. Path A — Windows GUI (recommended for most testers)

This is the easiest path. You will install Mercatura Core, make one small configuration file, and then leave Mercatura Core running while it synchronizes.

2.1 Download and verify

1. Download the official Windows Mercatura Core testnet package from the Mercatura-Core release page.
2. Download or copy the published SHA256SUMS file/checksum for the same release.
3. Verify the package before installing. If the checksum does not match exactly, stop and do not run the file.
4. Install Mercatura Core using the normal Windows installer or extract the official ZIP if the release is distributed as a portable package.

2.2 Run once, then close it

5. Start Mercatura Core once. This creates the default Mercatura data directory.
6. If it opens on the wrong network or has no peers, that is expected at this stage.
7. Close Mercatura Core completely before editing the configuration file.

2.3 Create mercatura.conf

Press **Windows key + R**, paste the following path, and press Enter:

```
%LOCALAPPDATA%\Mercatura
```

Create a new text file named **mercatura.conf**. Make sure Windows did not silently name it **mercatura.conf.txt**.

```
testnet=1
listen=1
addnode=BOOTSTRAP_HOST:27778
```

Replace the placeholder. Change BOOTSTRAP_HOST to the official testnet node hostname or IP. Example format only:
addnode=203.0.113.10:27778

2.4 Start Mercatura Core again

8. Launch Mercatura Core normally.
9. Confirm the application is using TESTNET. Do not continue if you intended testnet but the application identifies another network.
10. Leave the program running. The first synchronization can take time; do not assume it is broken simply because progress is not instant.
11. Open the built-in information/console window and check that at least one peer connection appears.

2.5 Verify from the built-in console

Run these commands one at a time:

```
getblockchaininfo
getnetworkinfo
getpeerinfo
```

You are looking for: **chain = test**, a nonzero peer count, and a block height that increases toward the public testnet tip.

3. Windows: synchronization, wallet, and inbound connections

3.1 How to know synchronization is finished

- The reported block height matches the current public testnet Explorer or another trusted Mercatura node.
- The best block hash matches the same height on another independent node.
- The node is no longer rapidly downloading historical blocks.
- Peer connections remain established after restarting Mercatura Core.

3.2 You do not need a wallet to run a node

The Qt application includes wallet functionality, but the networking/validation role is separate. If you create a wallet, use it only with testnet addresses. Mercatura's normal native testnet receive addresses use the tmca Bech32m-style prefix. Do not copy a mainnet address into a testnet test.

3.3 Optional: let other nodes connect to you

Your node can work with outbound-only connections. To make it more useful to the public testnet, allow inbound TCP 27778. This requires two separate things: your PC firewall must allow the port, and your router must forward TCP 27778 to the computer running Mercatura Core.

12. Give the Mercatura computer a stable local/LAN IP address or DHCP reservation.
13. In your router, create a port-forward rule: external TCP 27778 → this computer's local IP, TCP 27778.
14. In Windows Defender Firewall, allow Mercatura Core or create an inbound TCP rule for port 27778.
15. Do NOT forward TCP 27775. That is the RPC port and should remain private.
16. Restart Mercatura Core and leave it online. Inbound peers may appear gradually.

CGNAT / mobile Internet. Some ISPs do not give you a directly reachable public IPv4 address. If port forwarding never works because you are behind carrier-grade NAT, do not panic. Your node can still make outbound connections and fully validate the chain.

3.4 Clean shutdown

Use the application's normal Exit/Quit command and wait for shutdown to finish. Do not repeatedly kill the process or power off the PC while the node is writing chain state.

4. Path B — Ubuntu desktop or manually started daemon

Use this path if you are comfortable with a terminal but do not need a permanent system service.

4.1 Qt desktop

From the directory containing the official Linux binaries:

```
./mercatura-qt -testnet -addnode=BOOTSTRAP_HOST:27778
```

If the application starts normally, verify peers and synchronization in the GUI or built-in console exactly as in the Windows section.

4.2 Headless daemon

From the directory containing **mercaturad** and **mercatura-cli**:

```
./mercaturad -testnet -daemon -addnode=BOOTSTRAP_HOST:27778
```

Check it:

```
./mercatura-cli -testnet getblockchaininfo
./mercatura-cli -testnet getnetworkinfo
./mercatura-cli -testnet getpeerinfo
```

Stop it cleanly:

```
./mercatura-cli -testnet stop
```

4.3 Default Linux data location

The default Linux Mercatura data directory is:

```
~/.mercatura
```

The testnet chain data and authentication cookie live in the testnet network subdirectory beneath the chosen data directory. The configuration file is **mercatura.conf**.

No public RPC. For a normal desktop node, you do not need to open RPC at all. **mercatura-cli** can use the local authentication cookie. Never put 27775 on a public firewall rule just to make the CLI work.

5. Path C — Ubuntu 22.04 always-on full node

This path is for a home server or VPS that should run 24/7. It uses a dedicated service account, a fixed data directory, localhost-only RPC, and systemd.

5.1 Install the official binaries

Download and verify the official Linux release first. After extracting it, copy `mercaturad` and `mercatura-cli` into `/usr/local/bin`. The exact archive filename may change between releases, so this guide intentionally does not hard-code an unverified package name.

```
sudo install -m 0755 mercaturad /usr/local/bin/mercaturad
sudo install -m 0755 mercatura-cli /usr/local/bin/mercatura-cli
```

5.2 Create the service user and directories

```
sudo useradd --system --home /var/lib/mercatura --create-home --shell /usr/sbin/nologin mercatura
sudo mkdir -p /var/lib/mercatura/testnet
sudo mkdir -p /etc/mercatura
sudo chown -R mercatura:mercatura /var/lib/mercatura
```

5.3 Create the configuration

Create `/etc/mercatura/mercatura.conf` with:

```
server=1
listen=1
prune=0
txindex=0
rpcbind=127.0.0.1
rpccallowip=127.0.0.1
addnode=BOOTSTRAP_HOST:27778
```

This is a normal archival full node, not an Explorer node. `txindex` is intentionally off because ordinary nodes do not need it. If this server will power the Mercatura Explorer, use the Explorer runbook instead and enable `txindex`.

5.4 Protect the config

```
sudo chown root:mercatura /etc/mercatura/mercatura.conf
sudo chmod 0640 /etc/mercatura/mercatura.conf
```

6. Ubuntu always-on node: systemd and firewall

6.1 Create the service

Create `/etc/systemd/system/mercaturad-testnet.service` with:

```
[Unit]
Description=Mercatura Core Testnet Node
After=network-online.target
Wants=network-online.target

[Service]
User=mercatura
Group=mercatura
Type=forking
ExecStart=/usr/local/bin/mercaturad -testnet -daemon \
  -datadir=/var/lib/mercatura/testnet \
  -conf=/etc/mercatura/mercatura.conf
ExecStop=/usr/local/bin/mercatura-cli -testnet \
  -datadir=/var/lib/mercatura/testnet \
  -conf=/etc/mercatura/mercatura.conf stop
Restart=on-failure
RestartSec=10
TimeoutStopSec=120

[Install]
WantedBy=multi-user.target
```

6.2 Enable and start it

```
sudo systemctl daemon-reload
sudo systemctl enable --now mercaturad-testnet
sudo systemctl status mercaturad-testnet
```

6.3 Firewall

On a public Ubuntu server, allow SSH and Mercatura testnet P2P:

```
sudo ufw allow OpenSSH
sudo ufw allow 27778/tcp
sudo ufw enable
sudo ufw status
```

Do NOT run this command: `sudo ufw allow 27775/tcp` — Mercatura RPC is intentionally local/private. The Explorer or local CLI can use localhost without exposing RPC to the Internet.

6.4 Check the node

```
sudo -u mercatura mercatura-cli -testnet \
  -datadir=/var/lib/mercatura/testnet \
  -conf=/etc/mercatura/mercatura.conf getblockchaininfo

sudo -u mercatura mercatura-cli -testnet \
  -datadir=/var/lib/mercatura/testnet \
  -conf=/etc/mercatura/mercatura.conf getnetworkinfo

sudo -u mercatura mercatura-cli -testnet \
  -datadir=/var/lib/mercatura/testnet \
  -conf=/etc/mercatura/mercatura.conf getpeerinfo
```

7. Troubleshooting: follow this in order

Do not change five things at once. Check the simplest cause first, then move down the list.

Problem: zero peers

- Confirm you replaced `BOOTSTRAP_HOST` with the current official testnet hostname/IP.
- Confirm the line ends in `:27778`, not `:27775`.
- Confirm the bootstrap node itself is online.
- Confirm your computer can make outbound Internet connections.
- Restart Mercatura Core after changing `mercatura.conf`.
- On Linux, check `getpeerinfo` and the debug log before changing firewall settings.

Problem: node says mainnet / wrong chain

Stop the node. Confirm `testnet=1` is in `mercatura.conf` or that you launched with `-testnet`. Do not keep using a data directory if you are unsure which network created it.

Problem: CLI cannot connect

- Make sure `mercaturad` is actually running.
- Use the same `-testnet` setting as the daemon.
- Use the same `-datadir` and `-conf` paths if you started the daemon with custom paths.
- Do not solve a local CLI problem by opening RPC to the Internet.

Problem: service fails on Ubuntu

```
sudo systemctl status mercaturad-testnet
journalctl -u mercaturad-testnet -n 100 --no-pager
```

Typical causes are a typo in `mercatura.conf`, incorrect file permissions, an invalid bootstrap address, or a binary that is not actually installed at `/usr/local/bin`.

Problem: port 27778 is not reachable from outside

The node can still synchronize with outbound peers. For inbound reachability, check the OS firewall, router port forward, local IP address, and whether the ISP uses CGNAT. Never substitute 2775 when troubleshooting inbound P2P.

Problem: I changed testnet genesis / installed a new incompatible testnet build

A changed testnet genesis creates a different chain. Stop the old node and follow the release-specific reset instructions. Do not mix an old-genesis data directory with a new-genesis public testnet.

8. Routine operation and maintenance

8.1 Once per day or whenever you check it

Check	What good looks like
Peer count	At least one stable peer; ideally several independent peers
Chain height	Matches or is close to the current public testnet tip
Best block hash	Matches independent peers at the same height
Disk	Enough free space remains for chain growth
Logs	No repeating fatal/consensus/database errors
Uptime	Node restarts cleanly after OS reboot or planned maintenance

8.2 Useful Ubuntu commands

```
sudo systemctl status mercaturad-testnet
sudo systemctl restart mercaturad-testnet
journalctl -u mercaturad-testnet -f
```

8.3 Planned shutdown

Use the GUI's normal Exit/Quit command or use `mercatura-cli stop`. Wait for the process to finish. A clean shutdown gives the node time to flush chainstate and wallet data.

8.4 Pruning and txindex

Ordinary Mercatura nodes may run pruned and do not need `txindex`. This guide uses an unpruned node because it is simpler for early testnet troubleshooting. Explorer infrastructure is different: it should be archival with `txindex` enabled. Do not enable `txindex` just because you are running a normal wallet/full node.

8.5 What a good public testnet node contributes

- Independent validation of MercaHash blocks and Mercatura consensus rules.
- Additional peer diversity across hardware, ISPs, countries and NAT/firewall environments.
- Block and transaction relay capacity.
- Real evidence that nodes can restart, reconnect, resynchronize and recover from temporary network interruptions.
- A broader network that is not dependent on a single Mercatura-operated VPS.

Keep testnet running after mainnet exists. A healthy project keeps testnet as a separate development and upgrade network. Mainnet later uses P2P 27777 and private RPC 27776, but it must have its own release, genesis, data, service and launch checklist.

9. Final checklist — if every box is true, your node is working

- ☐ I installed the official Mercatura Core testnet build and verified its checksum.
- ☐ My node is explicitly running testnet.
- ☐ I replaced BOOTSTRAP_HOST with a real official testnet peer.
- ☐ I have at least one peer connection.
- ☐ My block height is syncing toward the public testnet tip.
- ☐ My best block hash agrees with another independent node at the same height.
- ☐ I did not expose RPC port 27775 to the Internet.
- ☐ If I wanted inbound peers, I opened/forwarded TCP 27778 only.
- ☐ I know how to shut the node down cleanly.
- ☐ I know where to check peers, chain state and logs if something stops working.

Quick reference

Item	Value
Testnet P2P	27778/tcp — public
Testnet RPC	27775/tcp — localhost/private only
Mainnet P2P later	27777/tcp
Mainnet RPC later	27776/tcp — localhost/private only
Windows data dir	%LOCALAPPDATA%\Mercatura
Linux data dir	~/.mercatura (default)
Config file	mercatura.conf
Node binaries	mercatura-qt, mercaturad, mercatura-cli

The simplest success test. If `getblockchaininfo` says testnet, `getpeerinfo` shows peers, your height reaches the same tip as another independent Mercatura node, and RPC 27775 is not public, you have successfully brought up a Mercatura testnet full node.

Technical basis: Mercatura Testnet Launch & Mainnet Transition Runbook; current Mercatura Core source defaults and branding; and the validated public-testnet network identity. This guide intentionally avoids hard-coding a bootstrap IP or release archive name because those operational values may change between public testnet releases.